Sahaj Bhadja

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Summary

Software engineer with 2+ years building **backend systems**, **SQL data access**, tested releases, and AI-assisted tools, focused on **IBM i**, **RPG/SQLRPGLE**, **CLP**, and enterprise integration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*
- Built Azure-backed AI education platform with normalized SQL schemas, row-level isolation, and indexed dashboards, reducing latency from 1.2 seconds to 350 milliseconds for 500+ campus users across releases.
- Reworked FastAPI background jobs with explicit states, failure isolation, and SQL-backed status tracking, processing 1K+ AI jobs while tripling throughput under concurrent instructor workloads, tickets, and reviews.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*
- Implemented React and Flask grading workflows for uploads, rubrics, history, and model selection, supporting 150+ instructors and 10K+ daily requests through production REST APIs, dashboards, and fixes.
- Designed concurrent file-ingestion services with ThreadPoolExecutor, OCR, and multimodal AI review, reducing feedback turnaround 60% while keeping batch processing testable, observable, and responsive for instructors and QA.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Troubleshot large-scale API integrations using OOP patterns, SQL workflows, AWS Glue, and S3, improving response times 30% while reducing technical debt 20% across release cycles and defects.
- Engineered cross-platform microservices with Jenkins CI/CD, VM automation, and 20+ PyTest, Mockito, unit, and integration suites while tracking defects through Agile sprint delivery with QA teammates consistently.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*
- Developed React/Node resource-management workflows with REST APIs, WebSockets, and modular services, synchronizing 2M+ records across locations with 33% fewer conflicts during rollout cycles and review tickets consistently.
- Implemented CRM modules for authentication, resources, billing, leads, permissions, and notifications, using Jenkins CI/CD and Docker staging to cut deployment time 50% safely across releases and fixes.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Built backend APIs for booking, slot selection, reservation tracking, coupons, and payment-status updates across 50K+ records, delivering sub-200-millisecond responses for client-facing workflows and support reviews reliably remotely.
- Mapped 12+ integration points across Flask APIs, Firebase synchronization, webhooks, payments, and cross-platform delivery, reducing design errors 25% through early stakeholder architecture reviews and documentation handoffs collaboratively.

Skills

IBM i & Backend: IBM i, RPG, SQLRPGLE, CLP, DB2 for i, SQL, Stored Procedures, Java, Python, Node.js, Flask, FastAPI
Data & Integration: REST APIs, Webhooks, Batch Processing, SQL Data Access, Relational Databases, PostgreSQL, Azure SQL, MongoDB, Firebase
Quality & Delivery: Unit Testing, Integration Testing, PyTest, Mockito, CI/CD, Jenkins, GitHub Actions, Git, Defect Tracking, Bug Fixes
AI & Collaboration: GitHub Copilot, Claude Code, AI Code Review, Security Review, Agile/Scrum, Sprint Planning, QA Collaboration, Code Reviews, Documentation

Projects

IBMi ServiceBridge | *IBM i, RPG, SQLRPGLE, CLP, DB2 for i, Node.js, SQL*
- Designed an IBM i integration project concept with RPG service programs, SQLRPGLE data access, CLP job orchestration, DB2 for i tables, Node.js adapters, unit tests, and reviewed AI-generated changes.

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, schema exploration, query generation, and validation, improving data-access productivity through Java services, TypeScript modules, and Docker-based local deployment workflows.

Linguistix | *Python, AWS, Docker, CI/CD, REST APIs*
- Developed and deployed a containerized REST data-processing API on AWS with CI/CD releases, validation checks, and Docker packaging, achieving 87% accuracy and production-grade reliability for automated workflows.